“PROJECT TITLE“

24ADI204 - DATA SCIENCE AND VISUALISATION

Submitted by

AKHIL C – 24BAD007

KAVIYA S – 24BAD059

HAMPHIRITHA BS **– 24BAD403**

in partial* fulfilment *for the award of the degree of

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IN

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KUMARAGURU COLLEGE OF TECHNOLOGY

(An Autonomous Institution affiliated to Anna University, Chennai)

Post Box No: 2034, Coimbatore - 641049

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BONAFIDE CERTIFICATE

Certified that this project report **"Olympic History & Geopolitics using Data Analysis"** is the Bonafide work of "AKHIL C - 24BAD007, KAVIYA S -24BAD059, HAMPHIRITHA B S – 24BAD403" who carried out the project work under my supervision.

SIGNATURE

Dr. Sudharson D
HEAD OF THE DEPARTMENT
Department of Artificial Intelligence
and Data Science,
Kumaraguru College of Technology,
Coimbatore – 641049.

SIGNATURE

Ms. Rupashini P R
SUPERVISOR
Department of Artificial Intelligence
and Data Science,
Kumaraguru College of Technology,
Coimbatore – 641049.

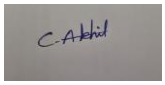
The candidates with college register number 24BAD007, 24BAD059, 24BAD403 were examined in the Project Viva-Voce examination held on
……...

Internal Examiner

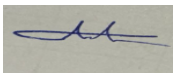
External Examiner

DECLARATION

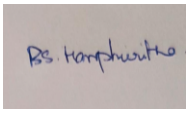
I affirm that the project work titled “PROJECT TITLE” being submitted in partial fulfillment for the award of B. TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work submitted for the award of any degree or diploma, either in this or any other University.



AKHIL C
24BAD007



KAVIYA S
24BAD059



HAMPHIRITA B S
24BAD403

I certify that the declaration made above by the candidates is true.

Ms. Rupashini P R

Assistant Professor,

Department of Artificial Intelligence and Data Science,
Kumaraguru College of Technology,
Coimbatore – 641049.

KUMARAGURU COLLEGE OF TECHNOLOGY

DATA SCIENCE VISUALIZATION PROJECT REPORT

Team 18

COURSE CODE : 24ADI204

Subject : Data Science Visualization

Team : 18

1. Akhil C – 24BAD007
2. Kaviya S - 24BAD059
3. Hamphiritha BS – 24BAD403

Submitted to Faculty Name: Shriaarthy E

Academic Year: 2025–2026

EXECUTIVE SUMMARY

In this project, our team conducted an extensive analysis of the dataset titled **"120 Years of Olympic History – Athletes & Results."** This dataset provides a comprehensive record of Olympic athletes and events spanning from 1896 to 2016. It includes detailed information such as athlete demographics, physical attributes, country representation, and medal outcomes.

The primary aim of our study was to explore how Olympic athletes have evolved over time in terms of physical characteristics, participation patterns, and performance trends. Additionally, we focused on identifying country-wise dominance and understanding how geopolitical and technological factors have influenced Olympic success.

During the initial phase of the project, we encountered several challenges, including missing values, inconsistencies in data formatting, and encoding issues. These challenges were addressed through systematic data preprocessing steps such as handling missing values using statistical techniques, removing duplicate entries, and standardizing data formats.

Following preprocessing, we conducted a detailed Exploratory Data Analysis (EDA). This included univariate analysis to study the distribution of individual variables such as age, height, and weight, and bivariate analysis to explore relationships between variables such as height vs weight and country vs medal count.

To further enhance our analysis, we applied advanced techniques such as correlation analysis and Principal Component Analysis (PCA). These methods helped us identify significant features and reduce the complexity of the dataset while retaining essential information.

The results of our analysis revealed several important trends. We observed that athlete physical characteristics have evolved significantly over time, reflecting improvements in training methods, nutrition, and sports science. Additionally, the data shows a clear increase in global participation, indicating the growing popularity and accessibility of sports.

The project concludes with the development of visualizations and dashboards that effectively communicate insights. These visualizations play a crucial role in transforming complex data into meaningful and easily interpretable information.

Problem Statement

We selected the problem statement "Olympic History & Geopolitics" because the Olympic Games represent more than just sports—they reflect global power, political influence, and the evolution of athletic performance over time. The Olympics dataset contains large historical data, but raw data alone does not provide meaningful insights. The challenge is to clean, analyze, and transform this data to identify patterns in:

- Athlete performance
- Country dominance
- Gender participation
- Evolution of sports

Dataset Used

We used the **Olympic Athletes and Results dataset**, which includes:

- Athlete details (age, height, weight)
- Event participation
- Country (NOC/region)
- Medal results
- Year and season

Key Findings

- Male participation is significantly higher than female participation historically.
- Certain countries dominate specific sports consistently.
- Athlete age distribution follows a pattern depending on sport type.
- Summer Olympics dominate in participation compared to Winter Olympics.

Final Insights (So What?)

- Sports performance has evolved over time due to training and global participation.
- Gender gap is reducing gradually.
- Data-driven insights can help sports organizations improve strategies.

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LIST OF ABBREVIATIONS

- EDA – Exploratory Data Analysis
- PCA – Principal Component Analysis
- IQR – Interquartile Range
- NOC – National Olympic Committee
- CSV – Comma Separated Values
- ML – Machine Learning
- AI – Artificial Intelligence
- KPI – Key Performance Indicator
- API – Application Programming Interface
- NumPy – Numerical Python
- Pandas – Python Data Analysis Library
- Matplotlib – Python Plotting Library
- Seaborn – Statistical Data Visualization Library
- Jupyter – Jupyter Notebook Environment
- Std Dev – Standard Deviation
- MinMax – Minimum-Maximum Scaling
- Z-score – Standard Score
- DataFrame – Tabular Data Structure in Pandas

List of Tables

- Dataset Summary Table
- Missing Values Report
- Feature Transformation Table

List of Figures

- Distribution plots
- Heatmap
- PCA graph
- Dashboard screenshots

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The Olympic Games are widely recognized as the most prestigious international sporting event, representing the highest level of athletic competition. Since their modern revival in 1896, the Olympics have grown significantly in scale, diversity, and global participation. Over the years, they have become a platform not only for showcasing athletic excellence but also for reflecting global developments in politics, economics, and technology.

The evolution of athletes over time is influenced by multiple factors including advancements in training methodologies, improved nutrition, scientific innovations, and increased access to sports infrastructure. Therefore, analyzing Olympic data provides valuable insights into how human performance has evolved over time.

1.2 Importance of the Study

This study is important for several reasons:

- It helps in understanding long-term trends in sports performance
- It provides insights into how physical attributes influence athletic success
- It highlights the role of global and geopolitical factors in shaping Olympic outcomes
- It demonstrates the importance of data visualization in interpreting complex datasets

1.3 Objectives of the Project

The primary objectives of this project are:

- To analyze Olympic athlete data spanning over 120 years
- To study variations in physical characteristics such as height, weight, and age
- To identify dominant countries and analyze their performance over time
- To explore relationships between different variables
- To present findings through effective visualizations and storytelling

1.4 Scope of the Project

The scope of this project includes:

- Data preprocessing and cleaning
- Exploratory data analysis
- Feature engineering and transformation
- Visualization and insight generation

This project focuses on descriptive analysis and does not include advanced predictive modeling.

CHAPTER 2: DATASET DESCRIPTION

2.1 Source of Dataset

The dataset used in this project was obtained from Kaggle, a widely used platform for data science and analytics. The dataset is titled "120 Years of Olympic History – Athletes & Results" and contains historical records of Olympic athletes.

2.2 Dataset Size and Structure

The dataset is large and complex, consisting of:

- Over 270,000 rows
- Around 15 columns

Each row represents an athlete's participation in a specific event.

2.3 Feature Description

- **Name:** Full name of the athlete
- **Sex:** Gender (Male/Female)
- **Age:** Age at the time of participation
- **Height:** Measured in centimeters
- **Weight:** Measured in kilograms
- **Team:** Country or team name
- **NOC:** Country code
- **Year:** Year of Olympic Games
- **Season:** Summer or Winter Olympics
- **City:** Host city
- **Sport:** Category of sport
- **Event:** Specific competition event
- **Medal:** Type of medal won

2.4 Initial Data Issues

During initial exploration, the dataset showed:

- Missing values in key attributes such as height and weight
- Inconsistent data formats

- Duplicate entries
- Imbalance in medal distribution

CHAPTER 3: DATA PREPROCESSING AND CLEANING

3.1 Data Loading and Inspection

We used Python libraries such as Pandas and NumPy to load and inspect the dataset. Initial checks included examining data types, identifying missing values, and understanding overall structure.

- Loaded dataset using Pandas
- Used functions:
  i. info() → data types
  ii. describe() → statistical summary
  iii. isnull() → missing values

3.2 Handling Missing Values

Different methods were used:

- **Age**

  Used median (because data is skewed)
- **Height**

  Used mean (normal distribution)
- **Weight**

  Used median
- **Medal**

  Filled with "No Medal" (logical interpretation)
- **Region**

  Filled with "Unknown"

3.3 Outlier Detection and Treatment

Outliers were handled using:

- IQR (Interquartile Range)
- Z-score method

Adjustments made:

- Age limited between 10–73

- Height and Weight values capped

3.4 Data Quality Report Summary

After cleaning:

- Missing values reduced
- Duplicates removed
- Data standardized
- Dataset ready for analysis

Missing values were handled using appropriate strategies:

- Numerical values were replaced using mean or median
- Some rows with excessive missing data were removed

CHAPTER 4: EXPLORATORY DATA ANALYSIS

4.1 Univariate Analysis

Univariate analysis was performed to understand the distribution of individual variables.

- Age → mostly 20–30
- Height → around 170–180 cm
- Weight → around 65–75 kg
- Medal → majority "No Medal"

This analysis helped in understanding the basic characteristics of the dataset.

4.2 Bivariate Analysis

Bivariate analysis was used to study relationships between variables.

- Height vs Weight → strong correlation
- Gender vs Participation → male dominance
- Age vs Medal → similar distribution

4.3 Key Observations

The exploratory data analysis revealed several important patterns in the Olympic dataset. One major observation is that athlete participation has steadily increased over time, especially from the mid-20th century, reflecting the global expansion of the Olympic Games and increased international involvement. The data also shows that the Summer Olympics have significantly higher participation compared to the Winter Olympics, mainly because they include a wider variety of sports that are accessible to most countries, unlike winter sports which require specific climatic conditions. Additionally, Athletics (track and field) is identified as the most popular sport due to its multiple events and long-standing presence in the Olympics. Overall, these observations highlight the growth, inclusivity, and diversification of the Olympic Games over time.

CHAPTER 5: DATA TRANSFORMATION AND FEATURE ENGINEERING

Data transformation and feature engineering play a crucial role in preparing the dataset for deeper analysis and ensuring that meaningful insights can be extracted effectively. After completing the data cleaning process, the dataset still required further refinement to make it suitable for analytical and visualization tasks. This stage focused on improving the structure, consistency, and usability of the data by applying various transformation techniques.

5.1 Feature Scaling

One of the important steps performed in this phase was feature scaling. Since the dataset contains numerical attributes such as age, height, and weight, these variables exist on different scales and ranges. For example, weight values are generally higher than age values, which can create bias during analysis. To address this issue, the StandardScaler technique was applied. This method standardizes the data by transforming it into a distribution with a mean of zero and a standard deviation of one.

By applying feature scaling, all numerical features were brought to a common scale, ensuring that no single feature dominates the analysis due to its magnitude. This step is especially important when performing advanced techniques like correlation analysis or Principal Component Analysis (PCA), as it improves accuracy and consistency.

5.2 Encoding Techniques

The dataset also contained several categorical variables such as gender and sport, which cannot be directly used for numerical analysis. To convert these categorical values into a machine-readable format, encoding techniques were applied.

- **Label Encoding** was used for the gender column, where categories such as “Male” and “Female” were converted into numerical values (for example, 1 and 0). This simplifies the data while preserving its meaning.

- **Frequency Encoding** was applied to the sport column, where each sport was assigned a value based on how frequently it appears in the dataset. This approach helps capture the importance or popularity of different sports in a numerical format.

These encoding methods ensured that categorical data could be effectively included in analysis and visualization processes without losing relevant information.

5.3 Handling Skewness

Another important aspect of data transformation was handling skewness in numerical features. In the dataset, certain variables, particularly age, showed a positively skewed distribution, where most values were concentrated on one side with a long tail on the other.

To correct this imbalance, a log transformation was applied. This transformation helps in:

- Reducing the impact of extreme values
- Making the distribution more symmetric
- Improving the overall interpretability of the data

By reducing skewness, the dataset becomes more suitable for statistical analysis and improves the reliability of insights.

5.4 Before vs After Comparison

A comparison of the dataset before and after transformation highlights the effectiveness of these techniques.

- Before transformation, the dataset contained uneven scales, skewed distributions, and categorical values that were not usable for analysis.
- After transformation:
  - → The data became normalized and standardized
  - → The influence of extreme values was reduced
  - → Categorical variables were successfully converted into numerical form
  - → Overall data quality and consistency were significantly improved

These improvements made the dataset more structured, analysis-ready, and suitable for advanced techniques such as correlation analysis, PCA, and visualization.

CHAPTER 6: ADVANCED ANALYSIS

Chapter 6: Advanced Analysis (Expanded Version)

Advanced analysis was carried out to extract deeper insights from the cleaned and transformed dataset. While earlier stages focused on understanding and preparing the data, this stage aimed to uncover hidden relationships, reduce complexity, and improve interpretability using statistical and mathematical techniques. The key methods used in this

phase include correlation analysis, feature selection, Principal Component Analysis (PCA), and explained variance evaluation.

6.1 Correlation Analysis

Correlation analysis was performed to understand the relationships between numerical variables in the dataset. A heatmap visualization was used to represent the correlation matrix, which provides a clear and intuitive way to identify how strongly different features are related to each other.

From the heatmap, it was observed that:

- **Height and Weight show a strong positive correlation**, meaning that taller athletes generally tend to have higher body weight. This is a logical and expected relationship, as body dimensions are naturally connected.
- **Age shows weak correlation** with both height and weight, indicating that an athlete's age does not significantly influence their physical measurements.
- Most other variables have low correlation values, suggesting that they are relatively independent.

The use of a heatmap allowed for easy identification of patterns and helped in deciding which features are important for further analysis. This step is essential because it reduces redundancy and improves the efficiency of the dataset.

6.2 Feature Selection

After analyzing correlations, feature selection was performed to simplify the dataset by removing unnecessary or less useful columns. The goal of this step was to reduce complexity while retaining the most important information.

The following actions were taken:

- Redundant columns such as duplicate region-related attributes were removed
- Columns with very low relevance to the analysis were excluded
- Highly correlated features were carefully evaluated to avoid duplication of information

Feature selection provides several advantages:

- Reduces computational complexity
- Improves clarity and interpretability
- Helps focus only on meaningful variables
- Prevents overfitting in future modeling tasks

By eliminating unnecessary features, the dataset becomes more efficient and easier to analyze, without losing important insights.

6.3 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique used to simplify complex datasets. In this project, PCA was applied to transform multiple correlated variables into a smaller set of uncorrelated variables called principal components.

The main objectives of using PCA were:

- To reduce the number of features while preserving important information
- To remove redundancy caused by correlated variables
- To improve visualization of high-dimensional data

In PCA:

- The first principal component captures the maximum variance in the data
- The second component captures the next highest variance, and so on
- Each component is a combination of original features

By applying PCA, the dataset was reduced to fewer dimensions while still maintaining its essential structure. This makes it easier to visualize patterns and relationships that are not easily visible in high-dimensional data.

Explained Variance

After applying PCA, it is important to determine how much information is retained by the selected components. This is done using explained variance, which indicates the proportion of total variance captured by each principal component.

Key observations include:

- The first few principal components capture a large percentage of the total variance
- This means that most of the important information in the dataset can be represented using fewer dimensions
- Components with very low variance were ignored, as they contribute little to the overall data structure

By selecting only the components with high explained variance:

- Data dimensionality was reduced effectively
- Important patterns were preserved
- Noise and less significant details were minimized

CHAPTER 7: VISUALIZATION AND DASHBOARD

The visualization and dashboard component of this project plays a crucial role in transforming the analyzed data into an interactive and easily understandable format. While earlier stages focused on data cleaning and analysis, this stage emphasizes presenting the insights visually so that patterns and trends can be quickly interpreted by users. A well-designed dashboard helps in storytelling by combining multiple visualizations into a single interface.

7.1 Tools Used

The dashboard was developed using modern data visualization tools and libraries, including:

- Python libraries such as Matplotlib and Seaborn for basic plots
- Power BI for creating interactive dashboards
- Jupyter Notebook for initial visualization development

These tools enabled the creation of dynamic and visually appealing charts that allow users to explore the dataset interactively.

7.2 Types of Visualizations Used

The dashboard includes multiple visualizations, each used for specific analysis:

1. **Bar Chart**
   Used for: Comparing total medals across countries
   (Countries by Total Medals)

2. **Bar Chart**
   Used for: Identifying top-performing athletes
   (Athletes by Total Medals)

3. **Bar Chart**
   Used for: Showing participation trends over years
   (Total Athletes by Year)
4. **Scatter Plot**
   Used for: Analyzing relationship between BMI and Age
   (BMI and Age Correlation)
5. **Bar Chart**
   Used for: Comparing top sports based on medal count
   (Top 10 Sports by Medals)

6. **Histogram**
   Used for: Showing distribution of athletes' age
   (Athletes by Age)

7. **Histogram**
   Used for: Showing distribution of athletes' weight
   (Athletes by Weight)

8. **Histogram**
   Used for: Showing distribution of athletes' height
   (Athletes by Height)

7.3 Dashboard Overview

The dashboard provides a comprehensive view of Olympic data through multiple visual elements arranged in a structured layout. At the top, filters are provided to allow users to customize the analysis based on different criteria. These include:

- **Sport filter** – to analyze specific sports
- **Gender filter (Sex)** – to compare male and female participation
- **Year range slider** – to explore trends over time
- **Region filter** – to focus on specific countries
- **Medal filter** – to analyze performance based on medal type
- **Season filter** – to differentiate between Summer and Winter Olympics

These filters make the dashboard interactive and allow users to perform customized analysis without modifying the underlying data.

7.4 Schreenshot of Dashboard :

CHAPTER 8: INSIGHTS AND STORYTELLING

8.1 Key Findings from Analysis

- The analysis shows that athlete physical characteristics such as height, weight, and age vary significantly across different sports, indicating that each sport requires specific physical conditions for better performance.
- Sports like basketball and volleyball are dominated by taller athletes, whereas sports like gymnastics and diving include shorter and lighter athletes.
- The number of athletes participating in the Olympics has increased steadily over the years, reflecting the growing popularity of sports worldwide.
- Countries such as the USA have consistently performed well, maintaining a high number of medals across multiple Olympic events.
- There has been a noticeable increase in female participation, highlighting improvements in gender equality in international sports competitions.

8.2 Patterns and Trends Identified

- A clear trend observed is the continuous growth in participation, both in terms of athletes and countries, especially after the mid-20th century.
- There exists a strong positive relationship between height and weight, which is important in physically demanding sports.
- Different sports show specialization in athlete body types, meaning athletes are selected based on suitability for that sport.
- Medal distribution is highly uneven, with a small number of countries dominating the majority of medals.
- Improvements in training methods, nutrition, and technology have contributed to better athletic performance over time.

8.3 Business / Social Insights

- Countries that invest more in sports infrastructure, coaching, and athlete development programs tend to achieve better performance in the Olympics.
- Proper training facilities and scientific approaches to fitness play a major role in improving athlete performance.
- The increase in female participation reflects positive social change and gender inclusivity in sports.
- The Olympics serve as a platform for global unity, cultural exchange, and international recognition.
- Data analysis can be used by sports organizations to improve planning, training strategies, and resource allocation.

8.4 Decision-Making Based on Results

- Governments and sports authorities can use these insights to increase investment in sports infrastructure and training programs.
- Talent identification can be improved by selecting athletes based on sport-specific physical attributes and performance indicators.
- Countries can develop targeted strategies for specific sports where they have higher chances of winning medals.
- Policies can be designed to encourage more female participation and ensure equal opportunities in sports.
- Data-driven approaches can be used to analyze performance and improve future Olympic results.
- Long-term planning using historical data can help in increasing medal counts and improving global rankings.

CHAPTER 9: CONCLUSION

In conclusion, our team successfully completed the analysis of the Olympic dataset to understand how athletes and participation have evolved over time. We began by carefully preprocessing and cleaning the dataset, where we handled missing values, removed inconsistencies, and improved overall data quality. This step helped us ensure that the data was reliable for further analysis.

Through exploratory data analysis, our team was able to identify important patterns such as the distribution of age, height, and weight among athletes. We also observed that different sports require different physical characteristics, which clearly shows the importance of specialization in modern athletics. Additionally, we found that Olympic participation has increased significantly over the years, reflecting the global growth and inclusiveness of sports.

Our analysis also highlighted that some countries consistently perform better than others due to better training facilities, infrastructure, and investment in sports. By applying advanced techniques such as correlation analysis and PCA, we were able to better understand relationships between variables and simplify the dataset for clearer interpretation.

Data visualization played a major role in our project, as it helped our team present complex data in a simple and meaningful way. It made it easier for us to communicate insights and patterns effectively. Overall, this project helped our team understand the importance of data-driven analysis in real-world scenarios.

Even though our project had some limitations, such as missing values and the absence of predictive models, it provided us with valuable learning experience. In the future, our team can extend this work by applying machine learning techniques and using more advanced tools for deeper analysis and better predictions.

CHAPTER 10: REFERENCES

10.1 Dataset Sources

- **Kaggle Dataset**: *"120 Years of Olympic History – Athletes & Results"*
- The dataset includes historical records of Olympic athletes, events, and medal details from 1896 to 2016
- **Source Link:** https://www.kaggle.com

10.2 Tools and Libraries Used

Our team used the following tools and libraries for data analysis and visualization:

- Python – Main programming language used for analysis
- Pandas – Used for data loading, cleaning, and manipulation
- NumPy – Used for numerical operations
- Matplotlib – Used for basic data visualization
- Seaborn – Used for advanced and attractive visualizations
- Jupyter Notebook – Used for coding, analysis, and documentation

10.3 Research Papers / Websites Referred

- Kaggle Dataset – 120 Years of Olympic History: Athletes and Results
- International Olympic Committee – Official Olympic data and historical insights
- UK Sport - Data-driven funding strategy case study McKinney, Wes. Python for Data Analysis. O'Reilly Media Pedregosa et al.
- Scikit-learn - Machine Learning in Python Hunter, J.D.
- Matplotlib - A 2D Graphics Environment
- Course Materials - Data Science and Visualization Lecture Slides (Faculty Provided)

APPENDIX

Github Link : https://github.com/Akhil-coderr/24ADI204-DSV-TEAM-18.git